

## Measurement of Self Inductance by Maxwell Bridge

### AIM:

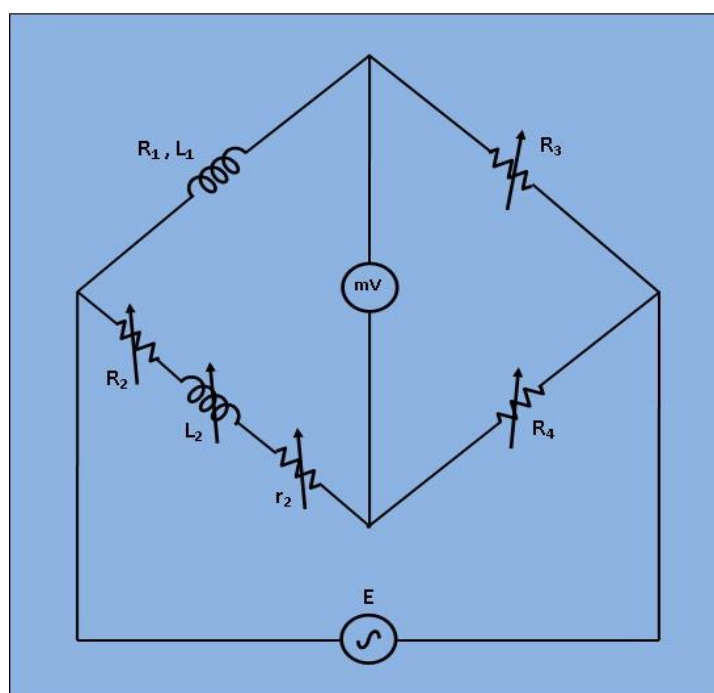
To determine the self-inductance of an unknown coil.

### APPARATUS REQUIRED:

- Virtual Labs

### THEORY:

This bridge circuit measures an inductance by comparison with variable standard self inductance. The connections for balance condition is shown in Fig. 1.



***Fig 1: Circuit Diagram for Measurement of Self Inductance by Maxwell Bridge***

Let,  $L_1$  = Unknown self Inductance of resistance  $R_1$ ,

$L_2$  = variable inductance of fixed resistance  $r_2$ ,

$R_2$  = variable resistance connected in series with inductor  $L_2$ ,

$R_3, R_4$  = known non inductive resistances,

At balance condition,

$$(R_1 + j\omega L_1) * R_4 = (R_2 + r_2 + j\omega L_2) * R_3 \dots (1)$$

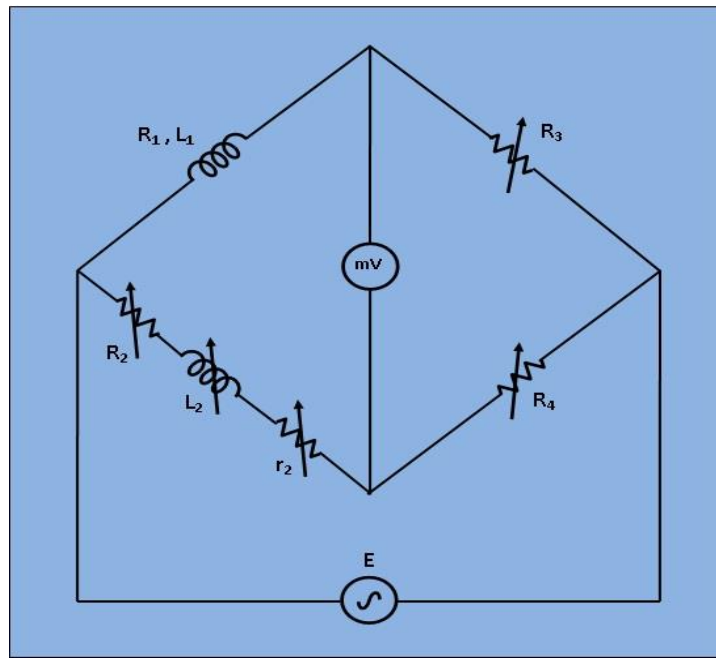
Equating both the real and imaginary parts in eq.(1) and separating them,

$$L_1 = \left(\frac{R_3}{R_4}\right)L_2 \dots (2)$$

$$R_1 = \left(\frac{R_3}{R_4}\right) * (R_2 + r_2) \dots (3)$$

Resistors  $R_3$  and  $R_4$  are normally a selection of values from 10, 100, 1000 and 10,000 $\Omega$ .  $r_2$  is a decade resistance box.

### **PROCEDURE:**



*Fig 1: Circuit Diagram for Measurement of Self Inductance by Maxwell Bridge*

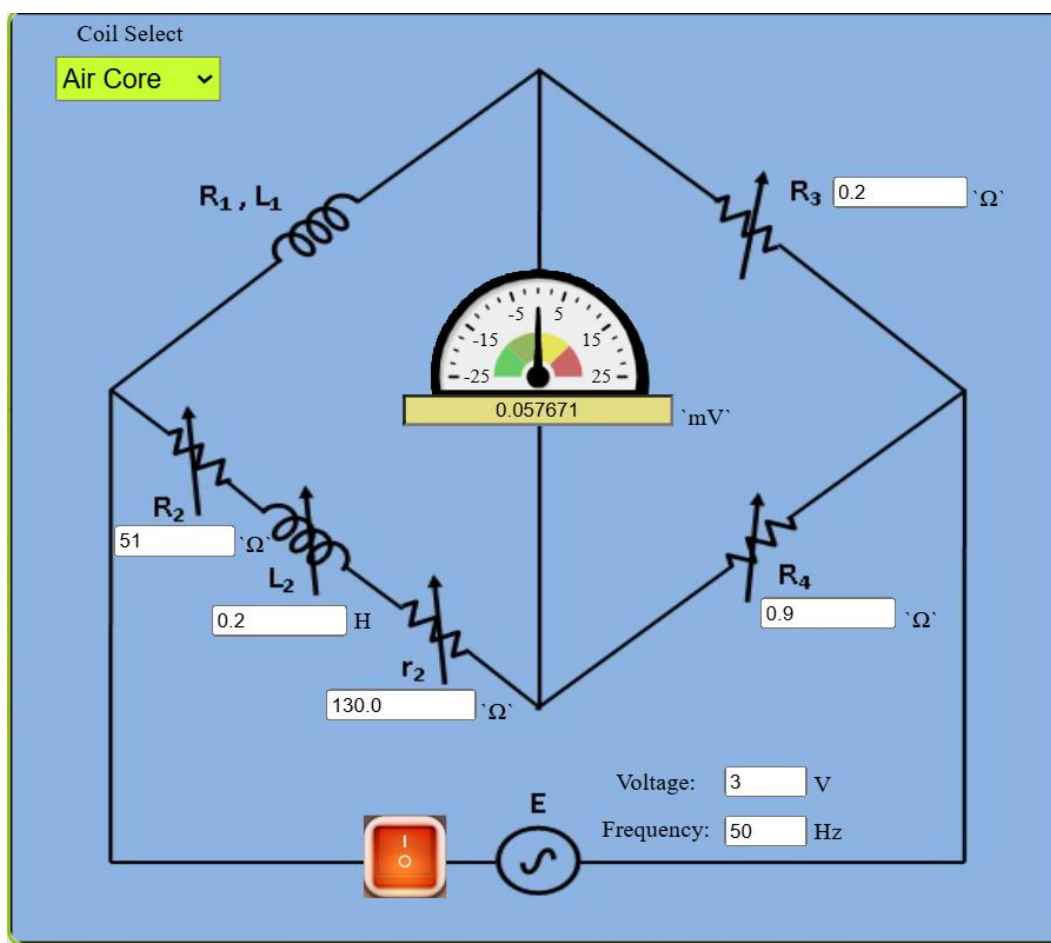
1. Apply Supply voltage from the signal generator with arbitrary frequency. (  $V = 3V$ ). Also set the unknown Inductance value from 'Set Inductor Value' tab.
2. Then switch on the supply to get millivoltmeter deflection.
3. Choose the values of  $L_2$ ,  $r_2$ ,  $R_2$ ,  $R_3$  and  $R_4$  from the inductance and resistance box. Vary the values to some particular values to achieve "NULL".
4. Observe the millivoltmeter pointer to achieve "NULL".

5. If "NULL" is achieved, switch to 'Measure Inductor Value' tab and click on 'Simulate'. Observe the calculated values of unknown inductance ( $L_1$ ) and its internal resistance ( $R_1$ ) of the inductor.
6. Also observe the Dissipation factor of the unknown inductor which is defined as

$$\frac{\omega L}{R} \text{ where } \omega = 2\pi f$$

## SIMULATION

### CASE 1: (AIR CORE)



The current voltmeter reading is: 0.057671 mv.

Now click on simulate to get:

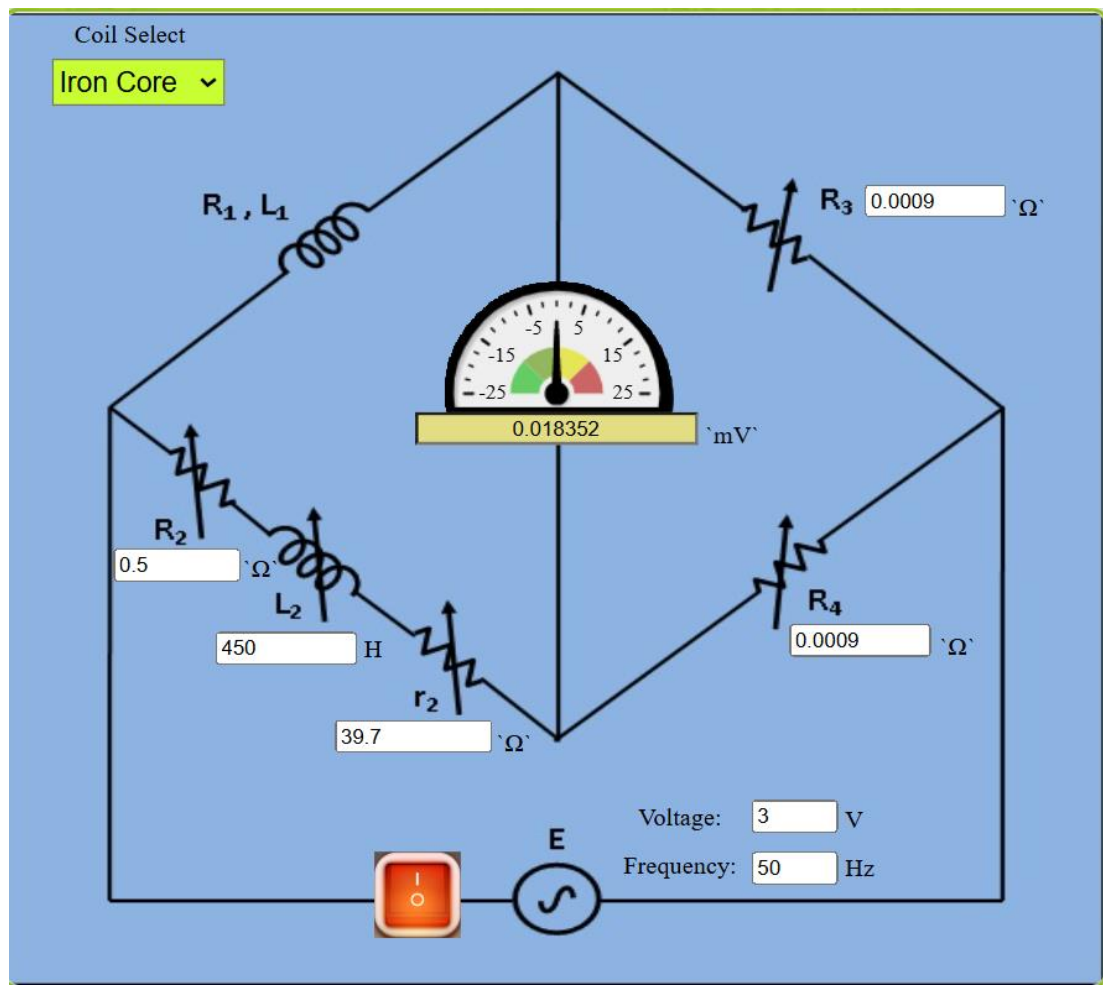
Inductor value (in mH): 44.444444444444

Resistance value (in Ohm): 40.222222222222

Quality Factor: 0.34696

Simulate

## CASE 2: (IRON CORE)



The current voltmeter reading is:  $0.018352 \text{ mv.}$

Now click on simulate to get:

Inductor value (in mH):  $449999.9999999999$

Resistance value (in Ohm):  $40.2$

Quality Factor:  $3514.9$

Simulate

## RESULT:

Thus, the unknown inductance is found using maxwell bridge in virtual lab.